# Bilingual ~~Retrieval-Augmented~~retrieval ~~Access~~augmented access to archived ANVISA ~~Regulatory~~regulatory ~~Documents: A Technical Evaluation Using Drug-Development and Health-Product Questions~~documents

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Abbreviations: AI, artificial intelligence; ANVISA, ~~Agencia~~Agência Nacional de ~~Vigilancia~~Vigilância ~~Sanitaria;~~Sanitária; AREE, equivalent foreign regulatory authority; BM25, Best Matching 25; CI, confidence interval; CLS, classification token; DDCM, Clinical Drug Development Dossier; DPI, Dossiê do Produto sob Investigação; DSUR, Development Safety Update Report; FDA, U.S. Food and Drug Administration; GMP, good manufacturing practice; ICH, International Council for Harmonisation; MRR, mean reciprocal rank; RAG, retrieval-augmented generation; RRF, reciprocal rank fusion; SUSAR, suspected unexpected serious adverse reaction.

## Abstract

Background. Regulatory teams often need to interpret long health-authority documents while working across languages. In Brazil, manyBilingual regulatory sourcesreview arerequires publishedcomplete insource-supported Portuguese,answers, whilenot companiesmerely and scientists may ask questions in English.citations.

Objective. To evaluate whether a bilingual retrieval-augmented generation prototype(RAG) couldagainst locate,archived cite,Agência andNacional summarizede selectedVigilância ANVISASanitária regulatory(ANVISA) requirements relevant to drug development and health-product regulation.documents.

Methods. Six ANVISAoverlapping regulatoryhistorical documentsstages wereproduced cleaned404 outputs (150 internal, 75 English repair, 75 Portuguese repair, 50 stress, 24 calibration, 30 post-calibration). Coauthor review covered 94 historical outputs and indexeda asseparate an40-answer article-levelassessment closed(20 corpus.bilingual Thepairs). systemAn was80-answer implementedrepair incomparison Pythonreceived 3.10author usingreview. LlamaIndex,Further sentence-transformers,comparisons rank-bm25,included Streamlit,120 coauthor-rated outputs with hybrid passages, BM25 passages or no retrieval, and the32 Google Generative AI interface. Densepositive-target retrieval used sentence-transformers/all-MiniLM-L6-v2; keyword retrieval used BM25; candidates were merged by reciprocal rank fusion and reranked with cross-encoder/mmarco-mMiniLMv2-L12-H384-v1. The final answer model was models/gemini-flash-lite-latest. The benchmark included 200 bilingual questions: 70 Portuguese answerable, 70 English answerable, 20 multi-article or exception questions, and 40 unanswerable or adversarial questions. An automated AI judge scored faithfulness, citation correctness, completeness, and refusal correctness on a 0 to 1 scale.

Results. Historical stress completeness averaged 0.943 (40 outputs; 95% pair-bootstrap interval 0.873 to 0.990). The finalcoauthor-reviewed article-awareassessment indexpassed contained10/20 382pairs detected(exact regulatory95% confidence interval [CI] 27.2 to 72.8%): 6/12 answerable, 0/4 partly answerable, 4/4 unanswerable. Author-reviewed repair success was 14/20 versus 17/20 (exact paired p = 0.375). Separate coauthor-rated success was 19/20 hybrid pairs (95% CI 75.1 to 99.9%), 18/20 BM25 pairs (68.3 to 98.8%) and 4/20 model-only pairs (5.7 to 43.7%). Hybrid minus BM25 success was +5 percentage points (95% pair-bootstrap interval -10 to +20). BM25 retrieved all required articles andin 481 chunks. In the first AI-authored prospective stress test, answerable30/32 questions hadversus source hit 1.000, source MRR 0.988, normalized article Recall@5 0.967, normalized article MRR@5 0.913, citation correctness 0.975, faithfulness 1.000, and completeness 0.943. Refusal correctness32/32 for unanswerable/adversarialhybrid questions was lower at 0.700. After refusal calibration using separate development data, a refusal-focused AI-authored test reached refusal correctness 0.944, with faithfulness 1.000, citation correctness 1.000, and completeness 0.908 on answerable controls.retrieval.

Conclusion. ARetrieved bilingualevidence RAG system may support regulatory information access whenimproved answers areon tiedthese toarchived verifiabletasks, articleswithout andestablishing refusalhybrid issuperiority evaluatedover asBM25. a primary endpoint. The projectCitations did not validateensure regulatorycompleteness advice.or Itapplicable testedlaw. aCurrent-law narrower use case: locatingaccuracy and summarizingfresh evidenceindependent insidevalidation aremain defined ANVISA corpus.unestablished.

## Introduction

Drug-developmentRegulatory decisionsdocument oftenreview beginbegins with practical questions: which provisions and exceptions must a narrowteam regulatorycheck question.when Aplanning a clinical-trial submission, preparing pharmacovigilance documentation or reviewing medical-device labelling? For example, a sponsor mayexamining needsuspension to know howof a Clinical Drug Development Dossier (DDCM) canwould beneed suspended,to whenlocate athe documentation, notification deadlines and implementation conditions in RDC 945/2024, Articles 85 and 89 [1]. Other review tasks include checking Development Safety Update Report (DSUR) mustsubmission be submitted, what minimum information makes anrequirements, adverse-event notification valid,elements whetherand aconditions foreign-manufacturedfor biologicalnon-printed productdevice caninstructions. beThese registeredexamples withoutmotivate registrationdocument review; they are not legal advice or evidence that this prototype answers current operational questions correctly. The suspension example uses RDC 945/2024 rather than Article 50 of the revoked RDC 9/2015 retained in itssome manufacturingarchived country,benchmark oranswers whether instructions for a medical device may be provided only in non-printed form. These questions affect trial operations, safety reporting, dossier planning, pharmacovigilance systems, and product labelling.[1,2].

TheA difficultybilingual isregulatory not only that the documents are long. The relevant answerteam may be in one article, while a nearby article discusses the same topic but does not answer the exact question. A user may also ask in English while the relevant source document is in Portuguese. InThe thatpractical setting,task extends beyond finding a generalrelated searchdocument: resulta is not enough. The userreviewer needs tothe knowapplicable whicharticle, documentits complete list of conditions and articleexceptions, supportan theanswer answer, and when the document collection does not containin the requested fact.language, and a clear statement of what the selected evidence cannot establish. An omitted notification category could leave a draft checklist incomplete; a refusal that suppresses supported information could obscure a requirement that warrants review. These are plausible consequences motivating the evaluation, not observed effects on submissions or patient outcomes.

Retrieval-augmented generation (RAG) systems may help by retrievingretrieves source passages before generating an answer [1,9].[3,4]. PriorA workstudy evaluatedof generativeU.S. AIFood and Drug Administration (FDA) guidance search overfound FDAthat guidancelocating documentsthe correct source did not guarantee a complete and reportedcorrect bothanswer promise[5]. The gap addressed here is whether article-level traceability is accompanied by complete task performance when Portuguese regulatory text must support answers in two languages, including questions whose requested details are only partly available. Retrieval accuracy, citation support, response language, evidence boundaries and limitationslegal [2].applicability Sourcetherefore citation was useful, but incomplete or incorrect answers remained a concern. That finding is directly relevantneed to regulatory affairs. A fluent answer can be unsafeexamined if it omits a condition, treats an exception as a rule, or answers a current operational question from a static regulatory document.separately.

This study evaluated that gap in a bilingual RAG prototype usingindexing selectedsix archived ANVISA documents. TheWe aim was not to build a regulatory-advice tool. The aim was to testasked whether the systemit could support traceable information access: retrieve the relevantcited documentprovision, preserve the requested requirements and article,exceptions, answer in Portuguese or English, cite the source,question's language, and refusedistinguish whensupported theinformation indexedfrom documentsunavailable detail. The evaluation also examined whether apparent repair gains survived question-overlap audits, selected human review and checks of reference scope. The study was designed to assess traceable access to a fixed document set; it did not provideevaluate theregulatory requestedadvice, information.live legal monitoring or decisions in an operational workflow.

ThisThe articleintended focusespractical oncontribution theis regulatoryevidence behaviorfor ofdesigning theand systemevaluating ratherassisted thandocument treatingreview. theOur

~~application~~experiments ~~as~~identify failure modes and evaluation checks for such a ~~generic~~workflow. ~~chatbot.~~They ~~The~~do ~~central~~not ~~questions~~demonstrate ~~were~~faster ~~whether~~review, ~~article-level~~fewer ~~retrieval~~submission ~~improved~~errors ~~traceability,~~or ~~whether~~readiness ~~bilingual~~for ~~queries~~autonomous ~~over Portuguese source text could be handled, whether generated answers preserved complete regulatory conditions, and whether the system could abstain when the answer was outside the closed corpus.~~use.

## Materials and Methods

Study design. This was a technical evaluation of a closed-corpus bilingual RAG prototype. The corpus was closed because the system was not allowed to search current web pages, submission portals, fee schedules, private company information, unpublished documents, or documents outside the local index. This design made the evidence boundary explicit. ~~If~~Unavailable detail required an explicit limitation; the ~~answer~~revised ~~was~~partial-answer ~~not~~endpoint ~~present~~also ~~in~~required ~~the~~any ~~indexed~~supported ~~documents,~~information. ~~the~~The ~~correct~~later ~~behavior~~model-only ~~was~~ablation ~~refusal.~~supplied no passages or web-search tool and permitted stored knowledge under a common prompt.

The evaluation was developed in stages. Initial tests identified problems in corpus cleaning, retrieval, English questions over Portuguese documents, answer completeness, citation behavior, and refusal behavior. Repairs were made on development data. The final reporting separates development runs, internal ~~held-out~~baseline and repair runs, the first AI-authored ~~prospective~~diagnostic stress test, and a later refusal-focused AI-authored test. ~~This separation is important because internal scores can be inflated when the same project team writes questions, reviews errors, and repairs the system.~~

Regulatory corpus. The corpus included six selected ANVISA documents covering clinical trials, biological products, medical devices, pharmacovigilance, and reliance-based review ~~[3-8].~~[1,2,6-9]. The selected documents ~~were chosen because they represent~~concern common ~~questions in~~ drug-development and health-product ~~work.~~topics; this selection did not establish that every indexed provision was in force at evaluation. The final corpus included IN 338/2024, RDC 406/2020, RDC 55/2010, RDC 751/2022, RDC 945/2024, and RDC 9/2015. Corpus composition and use cases are summarized in Table 1. RDC 9/2015 was already revoked by Article 98 of RDC 945/2024 when these evaluations were conducted; Anvisa identifies 1 January 2025 as the latter regulation's effective date. Its presence in the archived corpus and gold answers is a benchmark limitation, not a deliberately designed temporal-validity test. Article 90 of RDC 945/2024 contains submission-date transitional provisions; applicability cannot be inferred for a benchmark question without the relevant dates [1,2].

Each source file was checked against its document identifier and topic before final indexing. Source verification identified one local-file metadata mismatch: the local file name suggested stability studies, while the document content was RDC 406/2020 on pharmacovigilance. The final corpus table therefore used official regulation numbers and topics rather than local file names. Identifier and topic checks did not establish consolidated legal validity. A subsequent status audit checked official texts on 23 October 2026 and identified 18 additional constructed-benchmark rows citing wholly revoked articles of RDC 55/2010, alongside the 28 rows citing revoked RDC 9/2015 (46 of 200 rows, or 23 bilingual pairs). The RDC 55/2010 revocations follow RDC 875/2024 and RDC 948/2024 [10,11]. Consolidated texts also show amendments under IN 345/2025, RDC 777/2023 and RDC 967/2025 for IN 338/2024, RDC 751/2022 and RDC 406/2020, respectively. Supplementary Section S4 records the source checks and their limits; whole-article exposure counts do not certify the remaining gold answers.

SoftwareSystem and implementation. The prototype and evaluation scripts were implemented in Python 3.10. The environment file specified numpy 1.26, python-dotenv, PyTorch, sentence-transformers, ragas, Streamlit, LlamaIndex, llama-index-core, llama-index-embeddings-huggingface, llama-index-llms-google-genai, and rank-bm25 0.2.2 [10-15]. The user-facing application was a Streamlit app; a local launch and backend query smoke test are documented in Supplementary Figure S4. Indexing and evaluation were run from Python scripts in the project repository. The main implementation files were src/article_chunking.py, src/retrieval.py, src/prompts.py, src/engine.py, evaluation/publication_eval.py, evaluation/answer_quality.py, and evaluation/evaluation_statistics.py. Additional implementation and reproducibility details are provided in Supplementary Information and Supplementary Methods and Prompts.retrieval

The softwarePython implementation wasused derivedLlamaIndex, fromsentence-transformers, threeBM25 courseand notebooksStreamlit that[12-17]. progressedText fromcleaning retained document, chapter, section, article and citation metadata. Articles up to 500 cl100k_base tokens remained intact; longer articles were split with a local400-token LLMtarget scaffold,and to60-token overlap before heading context was added. Embeddings used all-MiniLM-L6-v2 with the recorded CLS-pooling wrapper and a chatbot,512-WordPiece tolimit. aBM25 simpleand RAGdense application.retrieval Theeach ANVISAreturned project100 keptcandidates; reciprocal-rank fusion used k = 60. A multilingual cross-encoder reranked candidates, and the teachingtop patternfive ofpassages asupplied Pythonanswer environment,context. configurationFigure module,1 modelsummarizes loader,the engine,workflow; Supplementary Section S1 records settings, prompts and appscoring interface,definitions. butImplementation replacedfiles, thechunking genericsettings, RAGindex designaudit with regulatory source cleaning, article-aware indexing, hybrid retrieval, reranking, bilingualand retrieval anchors,settings citation enforcement, refusal rules, and evaluation scripts. A recovered local ancestor of the ANVISA project was used to reconstruct the first working app and development decisions. Aare detailed decision trace is provided in Supplementary Table S5, with expanded development decisions and evaluation-stage disposition in Supplementary Tables S1, S3,S2, S3 and S4.S4, respectively.

TheEnglish answer-generationqueries modelused a deterministic 114-entry Portuguese glossary and thequestion-specific AI-judgequery, modeldocument and article anchors. Conditional neighboring-article and sibling-chunk expansion added context before generation. These rules were bothdeveloped models/gemini-flash-lite-latest.with exposure to benchmark failures; their evaluation uses development data. Article identifiers do not establish that every required clause reached the generator. The embeddingoriginal modelprompt required a same-language, article-cited answer from retrieved context and a fixed refusal when evidence was sentence-transformers/all-MiniLM-L6-v2.insufficient. TheGlossary rerankerexamples wasand cross-encoder/mmarco-mMiniLMv2-L12-H384-v1.fixed Therefusal local reproducibility package records the cache snapshot paths used during development, but model names rather than cache pathssentences are thelisted primaryin reproducibilitySupplementary identifiersTables becauseS5 localand paths are machine-specific.S6.

CorpusHistorical cleaninggeneration and article-awarejudging chunking.both Rawrequested regulatorythe textunpinned filesalias models/gemini-flash-lite-latest. Resolved July backend versions were cleanednot to remove markup residue, page boilerplate,preserved and formattingcannot artifactsbe whileinferred preservingfrom articlecurrent labelsdocumentation [17]. Later experimental responses recorded gemini-3.5-flash-lite. Historical timestamps, merge provenance and regulatorythe structure.distinction The final index used article-awarebetween chunking ratherand thanembedding arbitrarytoken fixed-sizeunits textare windows.documented Thein targetSupplementary unitSections was one regulatory article per chunk when possible. Chapter, section, document, source-file, article-number,S1 and citation-label metadata were retained with each chunk.S5.

ArticleAssessment chunking used a single-article maximum of 500 tokens. Longer articles were split into 400-token chunks with 60-token overlap. Embedding maximum length was 512 tokens. The final index audit recorded six source documents, 382 detected articles, 481 total chunks, 359 articles represented by a single chunk,sets and 23 long articles split into multiple chunks. Median chunk length was 155 tokens, maximum chunk length was 503 tokens, and no chunk exceeded 700 tokens. The closed-corpus workflow and index audit are shown in Figure 1.overlap

Retrieval,The reranking,constructed benchmark contained 200 question rows in 100 bilingual pairs: 140 direct-article, 20 multi-article/exception and answer40 generation.unanswerable/adversarial Retrievalquestions combined(Figure dense2). vectorThis searchconstruction total differs from the evaluated denominators. The internal assessment used 150 questions; each language-specific repair reused its 75-question internal subset with identical IDs and keywordtext. search.The Denseother retrievalstages usedcomprised 50 stress, 24 calibration and 30 post-calibration outputs. Together the sentence-transformerssix embeddingstages modelproduced [11].404 Keyword retrieval used BM25 over normalized text created from chunk contentoutputs and metadata254 fields,distinct includingquestion regulationIDs, identifier,without documentsemantic aliases,independence title,of citationthose label,IDs chapter,(Supplementary section,Section andS2). topicConstructed tagsbenchmark [12].composition Normalizationis removeddetailed accents,in lowercasedSupplementary text,Table and extracted alphanumeric tokens.S7.

The densestress retrieverfindings returnedinformed 100refusal candidates.calibration. BM25Three returnedsubstantively 100equivalent candidates.bilingual Candidatepairs listsrecurred werebetween mergedstress usingand reciprocalcalibration, rankcovering fusionthe current SUSAR form, device-notification fee and patient-identification premise; four further pairs had related content with kdifferent = 60, retaining up to 100 fused candidates. Candidate passages were then reranked with the multilingual cross-encoder reranker in batches of 8; this choice was based on multilingual passage-ranking models trained on mMARCO-style data [13].framing. The finalrepair, answercalibration contextand containedpost-calibration theresults toptherefore 5describe rerankeddevelopment passages.and cannot establish independent held-out performance or a causal calibration effect.

EnglishHistorical questions were answered in English, but retrieval was allowed to include conservative Portuguese anchors because the source documents were in Portuguese. Anchors included clinical trial/ensaio clinico, sponsor/patrocinador, biological product/produto biologico, medical device/dispositivo medico, stability study/estudo de estabilidade,scoring and pharmacovigilance/farmacovigilancia.coauthor The original question remained visible to the answer model. These anchors were retrieval aids only; they did not add facts to the answer.review

TheAutomated answermetrics promptassessed requiredevidence support, citation correctness, completeness and refusal, using continuous scores and archived labels [12,18,19]. Historical confidence intervals resampled bilingual pair IDs together with 2,000 percentile-bootstrap draws (seed 20260619). They are conditional on the modelarchived tolabels, answersources usingand onlyjudge retrievedand context,exclude answeruncertainty infrom thelegal-status or labeling errors. The same languagemodel asfamily the question, cite each material statement using the retrieved article-level citation label,generated and refusejudged ifanswers, retrievedlimiting evidenceindependence. didAudit notflags, containbootstrap enoughsettings, informationstage to answer the exact question. The refusal sentence was fixed by language: "The retrieved documents do not provide enough information to answer" for English questionsdenominators and "Oshistorical documentosestimates recuperadosare nao fornecem informacao suficiente para responder" for Portuguese questions. The final prompt is providedgiven in Supplementary MethodsTables S8, S9, S10 and Prompts.S11, Row-level generated answers, retrieval fields, and error flags are provided in Supplementary Data S1; AI-judge criteria, prompt, row-level scores, rationales, and stage-level conclusions are provided in Supplementary Data S2.respectively.

Benchmark.On 26 October 2026, H.L.R.J. reviewed and approved 94 selected historical outputs, including all 44 originally unanswerable stress/calibration/post-calibration outputs, 40 originally answerable outputs and ten unanswerable internal outputs. The mainratings benchmarkbegan

containedas 200an bilingualAI-prepared questions:draft; 70the Portuguesecoauthor answerablehad questions,seen 70some Englishscores, answerable questions, 20 multi-articlereferences or exceptioncoordinator questions,information. The review was author-led and 40unblinded. Completeness agreement used a judge threshold of 0.80, with 1.00 as sensitivity; refusal comparisons used the coauthor-defined unanswerable orstratum. adversarialOriginal questions. The benchmark had 100 Portuguese rowsratings and 100subsequent Englishanswerability rowsadjudications organizedwere asretained bilingualseparately pairs.(Supplementary TheSection auditS3). recordedReview 78 unique gold article citations. Benchmark structuresampling and goldauthor/judge source coverageagreement are shownreported in FigureSupplementary 2.Tables S12 and S13.

**ForFrozen answerable40-answer questions,author-led the benchmark recorded expected source documents, gold article labels, gold citation labels, expected answer terms, and a short expected answer. For unanswerable questions, the expected behavior was refusal. The benchmark included direct article questions, long-list questions, rule-plus-exception questions, false-premise questions, current operational questions outside the corpus, and adversarial questions that were regulatory in topic but not answerable from the indexed documents.assessment**

AutomatedTwenty AIbilingual judgepairs (40 questions: 12 answerable, four partly answerable and four unanswerable pairs), references and scoring rules.rules Automatedwere judgingfrozen wasbefore generation on 30 October 2026. The original six-document index and prompt produced one answer per question with gemini-3.5-flash-lite at temperature 0.2, without retries, repairs or new automated judging. H.L.R.J. rated every answer against the exact selected passages and frozen reference on 30 October, reporting no exposure to automated scores for this run. Pair success required both languages to meet the category rules and requested language. Partly answerable questions required supported information and identification of the unavailable detail; unanswerable items had no supported-content completeness endpoint. Exact 95% binomial intervals used becausepairs. anThe AI-authored, coauthor-approved questions followed earlier benchmark exposure; this is author-led assessment, not independent two-expertvalidation panel(Supplementary wasSection not available. This limits the claim level. The automated judge was used as a structured technical scoring tool, not as an independent regulatory expert. The scoring dimensions followed common RAG-evaluation concepts such as faithfulness or groundedness, context-supported citation, answer completeness, and refusal for insufficient evidence [10,16,17]. The judge received the question, system answer, expected answer, answerability flag, and retrieved evidence. It returned JSON scores from 0 to 1 for four dimensions: faithfulness, citation correctness, completeness, and refusal correctness, plus a short rationale. The exact prompt, model settings, row-level outputs, and stage-level recommendations are provided in Supplementary Data S2.S3).

**FaithfulnessDevelopment measured whether factual claims were supported by retrieved evidence. Citation correctness measured whether citations existed when an answer was given, referred to retrieved evidence,diagnostics and supportedexploratory the associated claim. Article-level citations were accepted when the supporting sentence was within a retrieved cited article; paragraph or item suffixes were not required. Completeness measured whether the answer covered the expected answer's essential information, including material conditions and exceptions. Refusal correctness measured whether an unanswerable question was refused without unsupported content, or whether an answerable question was answered rather than refused.repair**

TheRetrospective primaryretrieval reportedablations metricsused were160 continuousanswerable means,development questions in 80 bilingual pairs. They compared BM25, dense retrieval, fusion, reranking and rule removal on the fixed historical index. A window-pooling comparison changed only the embeddings of 35 overlength nodes; a targeted answer experiment used 30 affected questions and 20 seeded controls, three repeats per condition, totaling 300 generations and 300 same-model automated evaluations. Paired differences averaged repeats before resampling pairs with 10,000 draws. These reused-question analyses did not binaryselect experta adjudications.deployment Forsetting per-question(Supplementary failureSections review,S5 theand codeS6). flaggedRetrieval

answerableconditions rowsand aspaired problematic when the retrieved source was missing, article retrieval was incomplete, retrieved term coverage was below 0.50, generated answer-term coverage was below 0.50, citations were absent, cited labels were not among retrieved labels, or refusal occurred for an answerable question. AI-judge faithfulness below 0.80 was flagged as unfaithful. English completeness diagnosis used judge completeness below 0.80 to identify low-completeness rows for repair experiments. These thresholds were operational audit rules, not clinical or regulatory acceptance criteria. The resulting question-level flagscontrasts are provided with generated answersspecified in Supplementary DataTables S1S14 and with judge rationales in Supplementary Data S2.S15.

EvaluationFollowing stagesthe 40-answer assessment, controlled development experiments varied source completeness, instructions, source headings, language cues and leakagereference control.scope. DevelopmentNegative runscontrols and failed repairs were usedretained. The coauthor separately reviewed 44 first-repeat sibling-context answers after exposure to identifyscores and repairconditions. failures.These Internalexperiments held-outdiagnose runsselected werefailures usedand todo checknot thecreate repairednew system on the curated benchmark. When internal results became high, an AI-authored prospective stressindependent test wasitems run(Supplementary onceSections to test whether the internal pattern generalized. That first AI-authored prospective stress test was treated as diagnostic prospective evidenceS6 and wasS7). notSibling-context usedoutcomes, directlymechanism forcontrols tuning.and Afterarchived-reference itdecisions identifiedare refusalsummarized calibrationin asSupplementary theTables mainS16, weakness,S17 refusaland rulesS18, were adjusted using a separate development set. A second refusal-focused AI-authored test was then used to evaluate the repair.respectively.

A separate 482-node index and bundled repair were frozen before question authoring. The firstrepair combined structural parsing corrections, source-occurrence article completion, English-to-Portuguese retrieval translation, a named-regulation filter, language cues, revised partial-answer instructions, an exact-subject evidence check and deterministic citation formatting. Twenty AI-authored prospectivebilingual stresspairs testcomprised contained12 50answerable, four partly answerable and four unanswerable pairs; screening against 452 historical questions organizedreplaced asone 25semantic bilingualduplicate pairs.before Itfreeze. includedExecution 40completed answerable20 questionstranslations and 1080 unanswerableanswers by 1 November 2026 with gemini-3.5-flash-lite at temperature 0.2, without judge or adversarialrepair questions.calls. Pair success required content, evidence-boundary, language and citation-label criteria. Raw-output and common-formatter sensitivities were retained. All 80 outputs received personal author review on 1 November: 44 unblinded corresponding-author confirmations and 36 coauthor-reviewed AI-prepared ratings. The firstcorresponding AI-authoredauthor prospectivereported runthat requiredH.L.R.J. artifactsaw recoveryno afterseparate outputscores, folders had been moved; a preserved 40-row checkpoint was merged with a regenerated 10-row unanswerable/adversarial block using the same frozen settings. No benchmark, retrieval, prompt, judge, threshold,conditions or result-drivenprior tuning edits were madefindings before thatfinalizing recovery.those The36. refusal-focusedInitial AI-authoredAI test contained 30 questions organized as 15 bilingual pairs, with 18 unanswerable questionsscores and 12author answerableoverlays controls.remain separate (Supplementary Section S8).

StatisticalPair handlingsuccess required both languages to satisfy the specified criteria. Exact binomial intervals used pairs. The repair comparison also used paired bootstrap differences and auditan trail.exact Source-leveldiscordant-pair retrievaltest. wasAll summarizedintervals byare conditional on the selected questions and rubric and omit uncertainty from source hitdependence, review exposure and sourceAI-draft MRR. Article-level retrieval was summarized by Recall@k, Precision@k, MRR@k, and nDCG@k after lowercasing, accent removal, and whitespace normalization of citation labels. For generated answers, the system recorded citation presence, article citation presence, retrieved-citation validity, answer-term coverage, refusal detection, and AI-judge scores. When confidence intervals were reported in supplementary files, pair-clustered percentile bootstrap resampling used bilingual pair ID as the cluster key, 2000 resamples, 95% confidence level, and seed 20260619.assistance.

~~For~~Post ~~reproducibility,~~hoc ~~each~~common ~~evaluation row exported the question ID, pair ID, language, category, answerability flag, gold source documents, gold citations, raw vector candidates, BM25 candidates, fused candidates, final reranked contexts, generated answer, extracted citations, refusal detection, deterministic metrics, AI-judge scores,~~prompt and ~~judge~~lexical ~~rationale.~~baseline ~~The repository also contains manual-review templates. They were retained as audit scaffolds, but they were not completed as a two-expert validation dataset.~~comparisons

~~Streamlit~~On ~~interface.~~2 ~~A~~November ~~Streamlit interface was built as a local user-facing prototype for interactive question answering. The app accepted a free-text regulatory question, passed it to~~2026, the ~~same~~existing ~~retrieval~~40 ~~and~~FV ~~answer-generation~~questions ~~pipeline used in the scripted evaluation, and displayed the generated answer together with retrieved source passages and article-level citation labels. The interface was~~were used for ~~runtime~~80 ~~smoke~~new ~~testing~~answers with gemini-3.5-flash-lite at temperature 0.2. Identical neutral instructions permitted stored knowledge without passages; only the evidence block differed. RAG used saved integrated-repair contexts. Neither condition used web search. Expanded FV_012 scope included archived Articles 17, 30 and ~~demonstration~~31 before generation. Content and evidence-boundary success required complete supported information, no material unsupported claim, appropriate handling of unavailable detail and the ~~intended~~requested ~~workflow,~~language; ~~not~~appropriate ~~as~~abstention ~~a~~passed ~~separate~~absent-detail ~~evaluation~~items. ~~dataset.~~Citations ~~A~~were ~~screenshot~~secondary. ~~of~~Codex ~~a~~assessed ~~cited~~anonymized outputs with limited masking. On 2 November, H.L.R.J. personally drafted and approved ratings for all 118 distinct answers representing 120 condition records, including the subsequent BM25 extension. He reported no prior AI scores, condition identities or findings for these outputs, but had project/question exposure and answer style could reveal conditions. Ratings used common references rather than generator contexts. AI and ~~retrieved~~author ~~sources~~ratings ~~is~~remain ~~provided~~separate. ~~as~~Whole-pair ~~Supplementary~~bootstrap ~~Figure~~intervals ~~4.~~used 20,000 draws (seed 20261102).

The separate lexical baseline used the same 482 chunks, saved Portuguese retrieval queries, named-regulation filter and five seed chunks as the frozen hybrid/reranked retriever. BM25 used k1 = 1.5, b = 0.75 and epsilon = 0.25, with no dense branch or reranker. Thirty-two answerable or partly answerable language items had positive gold article targets; eight absent-detail items were excluded from recall. Both methods used the same full-article and adjacent-article completion policy, with variable final context lengths (Supplementary Section S10).

## Results

### ~~Corpus~~Historical development and ~~benchmark~~diagnostic ~~construction~~performance

The ~~final~~six-document index contained 382 parsed source records, including six ~~ANVISA~~preambles ~~documents,~~and ~~382~~repeated ~~detected regulatory articles,~~labels, and 481 ~~article-level~~chunks; ~~chunks~~359 ~~(Figure~~records ~~1).~~remained ~~Most~~intact ~~articles~~and 23 were ~~short~~split. ~~enough~~The ~~to~~benchmark ~~be~~covered ~~represented~~78 ~~by~~unique ~~one~~gold ~~chunk.~~article ~~Long~~citations. ~~articles~~RDC ~~were~~9/2015 ~~split with overlap to preserve continuity. This mattered because regulatory questions often ask for all items~~appeared in ~~an~~28/200 ~~article~~question ~~or~~rows. ~~for~~An ~~a~~additional ~~condition~~18 ~~that~~rows ~~appears~~cited ~~after~~wholly ~~the~~revoked ~~main~~RDC ~~rule.~~55/2010 articles, giving 46/200 exposed rows in 23 bilingual pairs. These are exposure counts, not adjudicated current-law error rates (Table 1; Supplementary Section S4).

Historical internal completeness averaged 0.808 across 120 originally answerable outputs. English completeness rose from 0.645 to 0.995 and Portuguese completeness from 0.970 to 0.995 on their reused repair subsets. The ~~200-question~~stress ~~benchmark~~assessment instead showed completeness 0.943 (95% pair-bootstrap interval 0.873 to 0.990; 40 answerable outputs), with refusal correctness 7/10 (0.700; interval 0.300 to 1.000; five bilingual pairs) under the original judge and labels. Later post-calibration refusal correctness was ~~balanced~~17/18 ~~by~~(interval ~~language~~0.833 to 1.000; nine pairs), with answerable-control completeness 0.908 (interval 0.792 to 1.000; 12 outputs). Different,

development-linked sets preclude a causal before/after claim. Original scores and ~~included~~stage ~~the~~denominators ~~intended~~are ~~mix~~retained ~~of~~in ~~regulatory~~Supplementary ~~question~~Data ~~types (Figure 2). Gold evidence covered 78 unique article citations. The source coverage was distributed across the six regulations, including clinical-trial, pharmacovigilance, biological-product, medical-device,~~S1 and ~~reliance~~S2. ~~questions.~~Historical ~~This~~estimates ~~structure~~and ~~was~~intervals ~~important~~are ~~because~~listed ~~high~~in ~~performance~~Supplementary ~~on~~Table ~~direct clinical-trial questions alone would not show whether the system handled rule-plus-exception or refusal boundaries.~~S11.

**~~Direct~~Human ~~article~~review ~~questions~~and answerability adjudication**

~~The system performed best when~~At the ~~question~~0.80 ~~matched~~judge ~~a~~threshold, ~~compact~~coauthor ~~article-level~~and ~~answer.~~automated ~~For~~completeness ~~example,~~classifications ~~it~~agreed ~~correctly~~on ~~answered~~38/58 applicable outputs (65.5%; exploratory pair-bootstrap interval 48.2 to 80.9%): 37 both passed, one both failed, 16 passed the judge but failed author review, and four showed the reverse. Among the 46 coauthor-classified answerable outputs, agreement was 37/46 (80.4%; 63.8 to 93.6%). Citation-support agreement was 44/44, an all-pass result that ~~a~~did ~~pharmacovigilance~~not ~~notification~~discriminate ~~under~~quality. ~~RDC~~These ~~406/2020~~targeted, ~~requires~~unblinded ~~identification~~agreement ~~of~~statistics ~~the~~do ~~notifier,~~not ~~identification~~estimate ~~of~~independent ~~the~~validation ~~patient, description of the adverse event, and name of the suspected medicine, citing RDC 406/2020, Art. 32. It also correctly handled a false-premise version of the same issue by answering no when asked whether a notification is valid without patient identification if the suspected medicine is named.~~accuracy.

~~The~~Archived-source ~~system~~adjudication ~~also~~reclassified ~~answered~~six ~~clinical-trial~~originally ~~conduct~~unanswerable ~~questions.~~outputs ~~When~~as ~~asked~~answerable ~~whether~~and ~~possible~~twelve ~~SUSAR~~as ~~reporting~~partly ~~allows~~answerable, ~~unblinding~~preserving ~~all~~the ~~participants,~~original ~~it~~labels. ~~answered~~Refusal ~~that~~of ~~unblinding~~an ~~should~~unavailable ~~be~~detail ~~limited~~was evaluated separately from completeness of supported information. In the retained stress, calibration and post-calibration unanswerable strata, author-rated appropriate refusals were 4/4, 10/10 and 12/12 outputs, or 2/2, 5/5 and 6/6 bilingual pairs. Exact pair-level 95% intervals were 15.8 to ~~the~~100.0%, ~~affected~~47.8 ~~participant~~to 100.0% and ~~cited~~54.1 ~~RDC~~to ~~945/2024,~~100.0%. ~~Art.~~The ~~67.~~twelve partly answerable outputs remained separate. These ~~examples~~small, ~~show~~differently ~~why~~composed ~~article-level~~strata ~~citation~~do ~~was~~not ~~part~~establish ~~of~~improvement ~~the~~(Supplementary ~~evaluation~~Section ~~rather than a display feature. The system's useful role was to narrow the search and provide a checkable summary; the regulatory user would still need to inspect the cited article before applying it.~~S3).

**~~Internal~~Frozen ~~development~~coauthor-reviewed ~~metrics~~answers**

~~The~~In ~~first~~the ~~held-out~~frozen ~~AI-judge~~author-led ~~evaluation~~assessment, ~~showed~~10/20 ~~high~~bilingual ~~faithfulness~~pairs met the full task criteria (50.0%; exact 95% CI 27.2 to 72.8%). Category results were 6/12 answerable pairs (50.0%; 21.1 to 78.9%), 0/4 partly answerable pairs (0%; 0 to 60.2%) and 4/4 unanswerable pairs (100%; 39.8 to 100%). At output level, 24/40 passed: 16/24 answerable, 0/8 partly answerable and 8/8 unanswerable. Full reference content was covered in 18/24 answerable outputs, but ~~weaker~~English ~~completeness.~~language ~~Among~~failures ~~120~~reduced ~~answerable~~complete-task ~~internal~~success. ~~held-out~~Requested ~~rows, mean AI-judge faithfulness~~language was ~~0.997,~~respected ~~citation~~in ~~correctness~~35/40 ~~was~~outputs: ~~0.965,~~15/20 English and ~~completeness~~20/20 ~~was~~Portuguese. ~~0.808.~~Table ~~Among~~2 ~~30~~and ~~unanswerable~~Figure ~~rows,~~3 ~~refusal correctness was 1.000. English answer completeness was~~report the ~~main~~paired ~~weak~~outcomes; ~~point~~Supplementary ~~in~~Section ~~this~~S3 ~~stage:~~provides ~~the~~all ~~English answerable subset had completeness 0.645. This result led to the English completeness repair work rather than to a claim that the system was already ready. The judged questions, scores,~~ratings and ~~rationales~~derived ~~for this stage are reported in Supplementary Data S2.~~results.

~~After~~The selected passages confirm missing list items in AV_004, AV_006 and AV_014, absence of the required Article 28 item in AV_013 Portuguese, and missing portions of the paired provisions in AV_016. AV_009 Portuguese correctly supplied the registration classes but could not supply the notification classes from its passages. All eight partly answerable outputs were blanket refusals. Five

omitted information that the coauthor judged supported by the selected passages; three lacked the required supported content in those passages. None identified the unavailable detail explicitly. The frozen prompt required a blanket refusal for partial evidence, whereas the prespecified partly answerable endpoint required a supported partial answer. This instruction and endpoint mismatch is a documented design limitation. AV_008 English ~~repair,~~cited supported pharmacovigilance statements about risk minimization instead of answering the ~~English~~contingency-plan ~~repair~~question. ~~set~~All 23 claim-bearing outputs had ~~mean~~supported ~~faithfulness 0.997, citation correctness 0.983, completeness 0.995,~~claims and ~~refusal~~citations ~~correctness~~according ~~1.000. After Portuguese residual-quality repair,~~to the ~~Portuguese~~coauthor, ~~repair~~illustrating ~~set~~why ~~had~~grounding ~~mean~~alone ~~faithfulness~~is ~~0.993,~~not ~~citation~~task ~~correctness~~success. ~~0.963,~~These ~~completeness~~failure ~~0.995,~~categories overlap and ~~refusal correctness 1.000. These results are shown as development history in Figure 3A. They should~~do not ~~be~~identify ~~interpreted~~causal ~~as~~effect ~~paired repeated-measure improvements because the stages used different evaluation sets. The row-level judge outputs and stage recommendations are reported in Supplementary Data S2.~~sizes.

~~**The saturated internal values, including refusal correctness of 1.000 on internal unanswerable rows, were investigated rather than accepted as evidence of a perfect system. They were considered potentially inflated because the benchmark**~~Retrieval **and ~~repair sets were created during development, targeted changes were made after error review, some metrics can saturate on small subsets, and the answer model and judge used the same model family. The first AI-authored prospective stress test was therefore run as a challenge to the internal pattern. Its refusal correctness of 0.700 showed that the internal 1.000 refusal result was too narrow for reliability claims. The AI-judge dossier documents this as a~~ diagnostic ~~recommendation rather than a tuning result, and the saturated-metric investigation is summarized in Supplementary Table S6.~~findings**

~~Completeness~~On ~~errors~~reused development questions, article Recall@5 was 0.6969 for citation-diversified BM25, 0.3344 for dense retrieval, 0.5687 for fusion and 0.9500 after reranking without anchors or neighbors. The full pipeline reached 0.9875. BM25 candidates with the full downstream rules also reached 0.9875, concealing one gain and one loss when dense candidates were added; the paired interval for that difference was -0.0187 to +0.0187. Thus, the comparison did not demonstrate an incremental mean benefit of the dense branch. These are retrieval endpoints, not answer-quality outcomes (Supplementary Section S5). Retrieval conditions and paired ablation contrasts are reported in Supplementary Tables S14 and S15.

~~Completeness~~The ~~errors~~300-answer ~~mattered~~window-pooling ~~more~~experiment ~~than~~changed ~~raw~~affected-stratum ~~retrieval.~~automated ~~Several~~completeness ~~errors~~from ~~occurred~~0.590 ~~event~~to ~~when~~0.617 ~~the~~(paired ~~correct~~difference ~~document~~+0.027; ~~or~~95% ~~article~~interval ~~was~~-0.027 ~~retrieved.~~to ~~In~~+0.099). ~~one~~Improvement ~~pharmacovigilance~~remained ~~question,~~unproven. ~~the~~Scores ~~system~~also ~~listed~~differed ~~the~~for ~~minimum~~byte-identical ~~elements~~prompts, ~~of~~and ~~a~~all ~~valid~~300 ~~notification~~automated ~~but~~faithfulness ~~omitted~~scores ~~the~~were ~~active-search~~1.000, ~~obligation~~including ~~when~~89 ~~required~~refusals. ~~information~~Selected ~~is~~passage ~~absent.~~completeness, ~~In~~source ~~one~~structure, ~~DSUR question, the answer gave the annual submission timing but omitted the ICH E2F condition. In one medical-device question, it omitted the requirement that non-printed~~language instructions ~~remain~~and ~~available~~question-reference ~~while~~scope ~~the~~each ~~product~~contributed ~~remains~~to ~~on~~case-level ~~the~~failures; ~~market.~~increasing ~~In~~context ~~a biological-product question, it gave the exception for a foreign-manufactured biological product but~~alone did not ~~first~~eliminate ~~state~~refusal ~~the~~of ~~general~~supported ~~rule.~~content. ~~In~~Detailed ~~a~~controls, ~~clinical-trial~~negative ~~question,~~results ~~it~~and ~~omitted~~exposure ~~a~~limits ~~30-business-day~~are ~~response~~in ~~deadline.~~Supplementary Sections S6 and S7.

~~These~~Exploratory ~~were~~repair ~~not~~on ~~cosmetic~~frozen ~~errors.~~author-led ~~In regulatory work, an omitted condition can change how a sponsor plans a submission, designs a safety process, or prepares product information. A system that retrieves the right article but omits the condition is useful only if the user still checks the citation.~~questions

Completed personal author review passed 14/20 baseline pairs (70.0%; exact 95% CI 45.7 to 88.1%) and 17/20 repaired pairs (85.0%; 62.1 to 96.8%). Four pairs improved and one worsened: difference 15.0 percentage points (pair-bootstrap 95% interval -5.0 to +35.0), exact two-sided p = 0.375. The ~~development~~coauthor ~~process~~failed ~~therefore~~repaired ~~changed~~FV_012 English (IRH_051) because the ~~evaluation~~technical-report ~~focus.~~requirement ~~Early~~is ~~work~~cited ~~asked~~to ~~whether~~Article ~~the~~17 ~~system~~rather ~~could~~than ~~find~~Article ~~the~~31 ~~right~~II. ~~source.~~Expanded ~~Later~~content ~~work asked whether the answer included all material elements from the source~~coverage and ~~whether~~citation ~~the~~support ~~system~~are ~~could~~distinct. ~~abstain~~Initial ~~from~~AI ~~unsupported~~scores ~~operational~~remain ~~claims.~~14/20 ~~This~~versus ~~sequence~~18/20 ~~is~~(difference ~~summarized~~20.0 ~~in~~points; ~~Table~~interval ~~2,~~+5.0 ~~with~~to ~~expanded~~+40.0; ~~failure-to-repair~~p ~~categories=~~ ~~in~~0.125). Initial AI raw-output and common-formatter results remain 14/20 versus 15/20 and 15/20 versus 18/20; these are historical sensitivities, not recomputed human endpoints (Table 3; Figure 4; Supplementary Section S8). Supplementary Table ~~S1.~~S19 ~~Representative~~separates ~~answer~~initial ~~outcomes~~AI ~~are~~assessments ~~provided~~from completed author review. The four gains comprised two answerable and two partly answerable pairs; the citation-support regression occurred in ~~Table~~one ~~3~~answerable pair. Thirteen pairs passed both conditions and ~~expanded~~two ~~in~~failed ~~Supplementary~~both, ~~Table~~so ~~S2.~~the net change was three successful pairs.

~~AI-authored~~Author-reviewed ~~prospective~~category ~~stress~~results ~~test~~were 10/12 versus 11/12 answerable pairs, 0/4 versus 2/4 partly answerable pairs and 4/4 unanswerable pairs in both conditions. Staffing and ruminant-documentation answers still refused supported information despite complete reference passages. Both conditions selected complete frozen-reference wording in all 32 positive language rows, so no retrieval-coverage gain was demonstrated. Mean context length increased from 5,737 to 16,487 characters. AI-draft assistance and differing review exposure limit inference. Supplementary Table S20 separates initial AI-assessed category counts from completed author review. The repaired answerable count is 11/12 under author review, compared with the initial AI-assessed 12/12; the partly answerable and unanswerable counts are unchanged.

The ~~first~~corresponding ~~AI-authored~~author ~~prospective~~confirmed ~~stress~~that ~~test~~broad ~~was~~FV_012 ~~more~~manufacturer/site ~~informative~~and ~~than~~GMP scope requires archived Articles 17, 30 VIII-IX and 31 I-III of RDC 55/2010. Only repaired English supplied the ~~internal~~expanded ~~post-repair~~content, but it failed the specific citation above. Expanded-scope author review yields 13/20 versus 17/20 pairs. This output-informed sensitivity preserves original references and AI scores. ~~It~~Article ~~contained~~30 ~~50~~is ~~questions,~~revoked; ~~including~~the ~~40~~adjudication ~~answerable~~concerns ~~rows~~the archived question (Supplementary Sections S4 and ~~10 unanswerable or adversarial rows. For answerable questions, source hit was 1.000 and source MRR was 0.988. Normalized article Recall@5 was 0.967 and normalized article MRR@5 was 0.913. Citation presence and retrieved-citation validity were both 0.975. AI-judge faithfulness was 1.000, citation correctness was 0.975, and completeness was 0.943.~~S8).

~~The~~Post ~~same~~hoc ~~test~~common ~~exposed refusal behavior as the main remaining weakness. Literal refusal detection for unanswerable/adversarial questions was 0.600,~~prompt and ~~AI-judge~~lexical ~~refusal~~baseline ~~correctness was 0.700. The main failure pattern was answering from nearby general regulatory text when the question asked for a more specific current or operational fact not present in the corpus. For example, the system answered a question about a current electronic SUSAR notification form model using the general electronic notification requirement, instead of refusing because the specific current form model was not in the indexed documents.~~results

~~Language-specific~~Coauthor-rated ~~prospective-test~~content and evidence-boundary success was 39/40 with hybrid passages, 38/40 with BM25 passages and 9/40 for the model alone, including eight appropriate abstentions per condition. Bilingual success was 19/20, 18/20 and 4/20 pairs (exact 95%

intervals 75.1 to 99.9%, 68.3 to 98.8% and 5.7 to 43.7%). Hybrid minus model-only success was +75 percentage points (descriptive pair-bootstrap 95% interval +55 to +90; 15 gains, no losses; exact paired p = 0.000061); hybrid minus BM25 was +5 points (-10 to +20; two gains, one loss; p = 1.000). This content endpoint differs from the earlier full-pipeline endpoint. Omissions affected 0/40, 1/40 and 31/40 answers; material claim flags affected 1/40, 1/40 and 14/40. Author-rated citation presence was 39/40, 38/40 and 0/40, and correction flags were 1/40, 2/40 and 32/40, without measured editing effort. Topic results ~~were~~are ~~similar~~in ~~but~~Supplementary ~~not~~Table ~~identical. Among answerable prospective-test rows, normalized article Recall@5 was 0.950 for English~~S21 and ~~0.983~~Supplementary ~~for~~Figure ~~Portuguese.~~S1 ~~AI-judge(~~Table ~~completeness4; was~~Figure ~~0.930 for English and 0.955 for Portuguese. AI-judge refusal correctness for unanswerable/adversarial rows was 0.800 for English and 0.600 for Portuguese. These results did not support a perfect-system claim. They supported a narrower claim: answerable regulatory questions were usually handled well, but refusal calibration remained a safety boundary.~~5).

~~Refusal~~In ~~calibration~~the matched retrieval baseline, all required articles appeared within five chunks for BM25 30/32 versus hybrid 32/32; full required article text after completion was present for 31/32 versus 32/32. Mean article Recall@5 was 0.958 versus 1.000 and MRR@5 was 0.932 versus 0.964. Pair-bootstrap difference intervals included zero. BM25 original-English queries found all required articles for 5/16 English questions, compared with 15/16 using the saved Portuguese translations; this language sensitivity is separate from the matched primary comparison (Supplementary Table S22).

~~After~~The BM25 extension used 39 new first-attempt calls and one exact-prompt cached answer, with saved hybrid comparators. Its English FV_016 omission coincided with missing Article 22 in context; its Portuguese FV_012 answer added intermediate-product manufacturers to the ~~first~~explicit ~~AI-authored~~name/address ~~prospective~~list. ~~stress~~The ~~test,~~hybrid ~~refusal~~failure ~~rule~~sequated ~~were~~DPI ~~revised~~with ~~using~~an ~~a~~Investigator's Brochure. Human and AI primary labels agreed on 117/118 distinct answers; one accepted model-only answer changed the AI count from 8/40 to 9/40 but not pair success. Secondary disagreements remain separate ~~refusal-development~~(Supplementary ~~set.~~Section ~~The revision instructed the answer model to treat related-but-incomplete context as insufficient and to refuse current forms, exact URLs, current fees or values, private contract terms, company-specific submission status, personal contact information, market prices, reimbursement, taxes, patents, future approvals, and advice not explicitly stated in the retrieved regulatory text.~~S10).

~~In the later refusal-focused AI-authored test, the system correctly refused 17 of 18 unanswerable questions by AI-judge scoring, giving refusal correctness 0.944. It did not over-refuse the answerable controls: answerable-control faithfulness was 1.000, citation correctness was 1.000, and completeness was 0.908. This improvement and the completeness trade-off are shown in Figure 3B. The result suggests that refusal calibration improved abstention behavior, but it also shows a cost: answerable-control completeness was lower than in the first AI-authored prospective stress test. The question-level judge rationales and remaining boundary failure are reported in Supplementary Data S2.~~

~~The remaining boundary problem involved questions where the corpus contained a related general rule but not the exact requested fact. This is a regulatory risk because related evidence can sound persuasive while still being insufficient. The safest interpretation is that refusal improved, not that refusal was solved.~~

**~~Summary of quantitative evaluation~~**

~~Across the development path, the main metrics changed in ways that affected the project interpretation. Initial internal held-out answer completeness was 0.808 overall for answerable rows, with English completeness 0.645. After targeted English repair, the English repair set reached completeness 0.995. After Portuguese residual-quality repair, the Portuguese repair set reached completeness 0.995. These internal repair results were not treated as independent proof because they followed error review and targeted changes.~~

~~The first AI-authored prospective stress test provided the main challenge to overfitting. It showed strong retrieval and answer grounding for answerable questions, with source hit 1.000, normalized article Recall@5 0.967, faithfulness 1.000, citation correctness 0.975, and completeness 0.943. It also showed weak refusal correctness at 0.700. The refusal-focused AI-authored test after separate calibration improved refusal correctness to 0.944 but had answerable-control completeness 0.908. Thus the main residual problem shifted from retrieval and language support to the boundary between partial related evidence and sufficient evidence.~~

## Discussion

~~Interpretation~~Traceable citations and high retrieval scores did not ensure task success. The coauthor-reviewed assessment passed 10/20 bilingual pairs despite support for claims and citations in all claim-bearing outputs (Table 2; Figure 3). Reviewers still need to check lists, exceptions, language and task relevance. Missing notification classes and truncated items could leave a draft checklist incomplete; the wrong-plan answer supplied supported information about the wrong task, and five English questions received Portuguese answers. These are implications of ~~the~~benchmark ~~development~~failures, ~~path~~not observed effects on regulatory work.

Partly answerable questions require both supported information and an explicit boundary around unavailable detail. The ~~main~~original ~~finding~~blanket-refusal ~~is~~instruction conflicted with that endpoint. Revised instructions and complete evidence improved selected development outcomes but retained failures and occasional task or formatting regressions. In the author-reviewed repair comparison, staffing and ruminant-documentation answers still withheld supported content. Four task-level gains coexisted with one citation-support loss and two persistent partial-answer failures (Figure 4). The difference was +15.0 points, with an interval from -5.0 to +35.0; reliable improvement was not ~~that~~established. ~~the~~Both ~~system~~conditions ~~achieved high average scores. The main finding is that a bilingual RAG prototype can support traceable access to~~already selected ~~ANVISA~~complete ~~documents~~frozen-reference ~~when~~wording, ~~it~~so ~~is~~retrieval-coverage ~~treated~~gains ~~as~~cannot ~~a~~explain ~~closed-corpus~~this ~~evidence-location~~comparison. ~~tool~~Refusal ~~rather~~assessment ~~than~~must ~~as~~address ~~a~~both ~~regulatory-advice~~invented ~~system. This interpretation is consistent with the broader RAG literature, which frames retrieval as a way to connect generation to external evidence,~~claims and ~~with~~suppression ~~prior~~of ~~regulatory~~supported ~~work showing that generative AI search can cite useful sources while still producing incomplete or incorrect answers [1,2,9].~~information.

The ~~figures~~common-prompt ~~support~~comparison ~~that~~supports ~~sequence~~supplying ~~rather~~evidence ~~than~~over ~~serving~~stored ~~only~~knowledge alone on these tasks (Table 4; Figure 5). Its expanded reference, new outputs and content endpoint prevent direct comparison with earlier repair scores. Appropriate abstention counts as ~~decoration.~~success ~~Figure~~on ~~1~~absent-detail ~~shows~~questions. BM25 nearly matched hybrid retrieval coverage and answer quality; the ~~engineering~~paired ~~turn~~interval ~~from~~does not establish superiority or equivalence. Different failures matter: BM25 omitted evidence and extended a ~~generic~~checklist, ~~RAG~~while ~~scaffold~~hybrid substituted the wrong dossier type. Coauthor confirmation strengthens assessment but does not remove development exposure. A strong lexical baseline is therefore necessary before attributing gains to ~~a closed, article-aware ANVISA pipeline. Figure 2 shows why the benchmark was not limited to easy article lookup: it mixed Portuguese and English questions, single-article questions, exception questions, and unanswerable or adversarial questions. Figure 3A then shows the development history after weak points were identified, while Figure 3B shows the later targeted repair of refusal behavior and the remaining completeness trade-off. Together, the figures make the main interpretation clearer: once article-aware indexing, hybrid retrieval, and reranking were in place, ordinary source~~ retrieval ~~was no longer the dominant weakness. The important question became whether the retrieved article was sufficient for the exact answer requested.~~complexity.

~~That~~Corpus ~~shift~~validity is ~~important~~a ~~for~~separate limitation. A targeted consolidated-text check retained the staffing criterion in RDC 406/2020 Article 18, but confirmed RDC 55/2010 Articles 22 and 30 were revoked by RDC 948/2024 [7,8,11]. The ruminant-documentation target and FV_012 Article

30 components therefore remain historical reading targets. Separate transmissible spongiform encephalopathy documentation rules require product-specific applicability review. Current-law corpus rebuilding, reference validation and model reevaluation remain incomplete; archived-text success must not be interpreted as current regulatory ~~use.~~accuracy ~~Metrics~~(Supplementary ~~showed~~Section ~~the overall pattern, but the examples explained the risk. The incomplete answers were not only stylistic omissions. They involved active-search obligations, ICH E2F conditions, market-availability requirements for instructions for use, the general rule before a biological-product exception, and a 30-business-day deadline. Similarly, refusal failures involved questions that sounded regulatory but asked for current operational facts such as a form model, phone number, petition code, or private case status. These examples show why completeness and refusal should be primary endpoints in regulatory RAG evaluation, not secondary notes.~~S4).

~~Use~~Fresh ~~in~~independent ~~drug~~validation remains absent. AI-authored questions, development exposure, post hoc analyses, source dependence and ~~regulatory~~single ~~affairs~~generations limit inference. Historical generation and judging shared an unresolved alias; window-pooling scores used the generation model and lack complete human verification. The new comparison has coauthor assessment with prior question exposure and limited condition masking; this is not independent external validation. BM25 generation was sequential and reused saved hybrid comparators. The English embedding model was not compared with a multilingual replacement [20,21]. Repeated generations do not create independent questions.

~~The most realistic use of this prototype is assisted navigation of a defined document set.~~ A ~~regulatory~~candidate ~~user~~workflow ~~could~~would ~~ask~~present ~~a~~the question, ~~receive~~answer, ~~a~~supporting ~~short answer,~~passages and ~~inspect~~unavailable details for professional verification against the ~~cited~~official ~~article~~current ~~before~~text. ~~using~~The ~~the information. This could reduce time spent searching across long Portuguese documents, especially for international teams working in English. It should not be used as a stand-alone regulatory decision tool because it~~prototype does not ~~access~~maintain ~~current~~consolidated ~~portals, fee schedules, forms, company-specific status, authority correspondence,~~law or ~~there~~liably ~~full~~warn ~~current~~of ~~legal~~superseded ~~context.~~provisions. This study evaluates model answers and retrieval, not superiority over web search or professional workflow benefit. No review time, correction effort, decision quality or operational safety was measured. Future workflow testing would require frozen current-law references and realistic tasks.

~~In practical use, the Streamlit prototype illustrates how the system could function as a lightweight evidence-navigation app rather than a regulatory decision engine. A regulatory affairs user could enter a question during dossier preparation, clinical-trial planning, pharmacovigilance review, or medical-device labelling work, then inspect the cited ANVISA article before using the answer. This use case is most appropriate for locating and summarizing evidence already present in the indexed corpus. It is not appropriate for current portal instructions, company-specific submission status, fee values, legal advice, or final regulatory strategy without human verification.~~

~~The examples map to ordinary development and submission work. Before or during a clinical trial, a sponsor may need to check DDCM suspension procedures, DSUR timing, or the boundary between SUSAR reporting and trial unblinding. During safety-system design, the team may need to confirm the minimum information that makes a pharmacovigilance notification valid and what follow-up is expected when information is incomplete. During product-registration planning, a biological-product team may need to understand whether an exception applies or whether the general manufacturing-country registration rule still controls the case. For medical devices, labelling and supply teams may need to know whether non-printed instructions for use are acceptable for the intended use setting.~~

~~Confidence should therefore depend on the question type. Confidence is highest for direct article questions, definition-like questions, and false-premise questions where the cited article clearly supports the answer. Confidence is moderate for rule-plus-exception questions, multi-article questions, and complete-list questions, because these are the settings where the evaluation found omissions. Confidence is low for current operational details, exact portal codes, current MedDRA versions, phone numbers, URLs, fee information, product-specific strategy, private case status, legal~~

advice, or future approval likelihood. For those questions, the safest behavior is refusal or a statement that the indexed documents do not provide enough information.

**Limitations and future work**

This study used a selected closed corpus and does not cover all ANVISA rules or current operational resources. Benchmark construction did not include an independent panel of regulatory experts. Automated judging was used and should be interpreted as technical scoring, not expert validation. This choice is consistent with current work on scalable LLM evaluation, but the literature also warns that automated judges may show bias and incomplete agreement with human judgment [10,16,17]. In this study, that limitation is especially relevant because the answer model and automated judge used the same Gemini model family.

The 1.000 values are therefore reported as part of the quality story, not as a claim of perfection. Internal refusal correctness reached 1.000, and some post-repair judge scores approached or reached 1.000, but those values came after benchmark construction, error review, and targeted repair. The first AI-authored prospective stress test was important because it challenged those saturated values without tuning from that set. It contradicted the internal refusal result: refusal correctness fell to 0.700 even while answerable-question retrieval and grounding remained strong. Figure 3A should therefore be read as development history, and Figure 3B as evidence that refusal calibration improved the main safety weakness but introduced a visible trade-off, with answerable-control completeness below the first prospective answerable set. The most defensible claim is limited but useful. The system can help locate and summarize evidence in selected ANVISA documents, provided users verify the cited article and treat refusal boundaries seriously.

Future studies should test the system with regulatory-affairs users and measure whether it reduces time to locate evidence without increasing overconfidence. A stronger validation study should include independent expert review of selected answers, blinded error classification, prospective question authoring by users not involved in system development, and comparison with standard document search. The corpus should also be expanded and versioned so users can see when a question requires current information outside the indexed corpus.

## Conclusion

AThis archived-corpus study addresses the gap between finding a regulatory citation and delivering a complete bilingual RAGanswer prototypeto cana helpdocument-review retrievequestion. Human review exposed omitted requirements, wrong-language and summarizewrong-task selectedanswers, ANVISAand regulatoryrefusal documentsof whensupported information. The contribution is an auditable set of failure cases and evaluation checks: preserve full provisions, verify scope and language, and distinguish unavailable detail from supported content. Coauthor-rated answers favored retrieved evidence over the answermodel isalone, presentbut inhybrid theretrieval indexedshowed corpusno established advantage over matched BM25 retrieval. Current-law accuracy and thefresh citedindependent articlevalidation isremain checked. The evaluation showed useful performance for clinical-trial, pharmacovigilance, biological-product, medical-device, and reliance questions. It also showed why regulatory RAG evaluation should report concrete successes and failures. The central safety boundary is not only retrieval accuracy; it is the ability to avoid incomplete answers and to refuse when the documents do not contain the requested fact.unestablished.

## Declarations

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Author Contributions: C.V.M.-P. conceived and designed the study, developed the software pipeline, prepared the corpus, constructed the benchmark, performed the analyses, interpreted the results, and drafted the manuscript. H.L.R.J. contributed to the regulatory and pharmaceutical framing,

interpretation of results, author review of questions and generated answers, and critical revision of the manuscript. ~~Both authors approved the final manuscript and agree to be accountable for the work.~~

Data Availability: ~~The~~Code, ~~code,~~curated ~~benchmark~~data, ~~manifests, generated~~frozen outputs, ~~prompts,~~separate AI and ~~non-restricted~~author ~~derived~~ratings, ~~artifacts~~and figure sources are archived ~~in~~as ~~a~~v0.2.0 at https://doi.org/10.5281/zenodo.23909097 and https://github.com/camontefusco/ANVISA_RAG_publication_repository_20260720/releases/tag/v0.2.0. The accompanying package adds these verified citations without changing scientific results. Historical outputs and original reviews remain separate from derived analyses. Full source/index corpora, credentials and private ~~project~~coordination ~~repository during manuscript preparation and will be made publicly available on GitHub after manuscript acceptance. Files needed for peer review~~keys are ~~included as supplementary materials. Official ANVISA source URLs, source-file checksums, and the cleaning script are provided in the source reproducibility folder. Full cleaned copies of ANVISA regulatory text are derived from public agency documents; if redistribution of full cleaned text is restricted or unnecessary, the public repository will provide official source links, checksums, and cleaning scripts rather than republishing the full source text.~~excluded.

Ethics ~~Approval:~~statement: ~~Not required because the~~The study used public regulatory documents and generated benchmark questions. Human assessment consisted of coauthor review of model outputs. No ~~human participants,~~ patient data, personal health data, animal ~~data,~~experiments or clinical ~~trial data~~trials were ~~used.~~involved.

AI assistance disclosure: OpenAI Codex (OpenAI) was used to assist software implementation and debugging, data and source-file audits, analysis-script preparation, manuscript drafting and revision, and preparation of documents and figures from inspectable project data. AI-generated benchmark questions, answers and judge outputs are described separately in Methods and the supplementary records. Historical assistant model versions and complete session-level usage metadata were not consistently recorded. The authors retain responsibility for the manuscript, software and interpretations; AI assistance does not constitute independent human verification. Human regulatory adjudication still pending during this revision is identified explicitly in the relevant records.

Clinical Trial Registration: Not applicable.

## References


1. LewisAgência P,Nacional Perezde E,Vigilância PiktusSanitária. A,Resolução etda al.Diretoria Retrieval-AugmentedColegiada GenerationAnvisa No. 945, de 29 de novembro de 2024. Guidelines and procedures for Knowledge-Intensiveclinical NLPtrials Tasks.for Advanceslater inmedicine Neuralregistration. InformationAnvisaLegis. ProcessingAccessed Systems.19 2020;33:9459-9474.July 2026.

2. ProestelAgência S,Nacional Jengde LJB,Vigilância SmithSanitária. C,Resolução etRDC al.No. Semantic9, Searchde of20 FDAde Guidancefevereiro Documentsde Using2015. GenerativeRegulation AI.for Therclinical Innovtrials Regulwith Sci.medicines 2025;59(5):1148-1159.in doi:10.1007/s43441-025-00798-8.Brazil. Revoked by RDC 945/2024, Art. 98, effective 1 January 2025. Historical corpus source. AnvisaLegis. Accessed 19 July 2026.

3. AgenciaLewis NacionalP, dePerez VigilanciaE, Sanitaria.Piktus InstrucaoA, Normativaet -al. INRetrieval-Augmented No. 338, de 29 de novembro de 2024. Establishes the list of equivalent foreign regulatory authorities and criteriaGeneration for optimizedKnowledge-Intensive reviewNLP underTasks. RDCAdvances 945/2024.in AnvisaLegis.Neural AccessedInformation 19Processing JulySystems. 2026.2020;33:9459-9474.

4. AgenciaGao NacionalY, deXiong VigilanciaY, Sanitaria.Gao ResolucaoX, deet Diretoriaal. ColegiadaRetrieval-Augmented - RDC No. 406, de 22 de julho de 2020. Good pharmacovigilance practicesGeneration for holdersLarge ofLanguage humanModels: medicineA registrations.Survey. AnvisaLegis.arXiv. Accessed2023. 19 July 2026.doi:10.48550/arXiv.2312.10997.

5. AgenciaProestel NacionalS, deJeng VigilanciaLJB, Sanitaria.Smith ResolucaoC, -et RDCal. No.Semantic 55, de 16 de dezembro de 2010. RegistrationSearch of newFDA biologicalGuidance productsDocuments andUsing biologicalGenerative products.AI. AnvisaLegis.Ther AccessedInnov 19Regul JulySci. 2026.2025;59(5):1148-1159. doi:10.1007/s43441-025-00798-8.

6. AgenciaAgência Nacional de VigilanciaVigilância Sanitaria.Sanitária. ResolucaoInstrução Normativa - RDCIN No. 751,338, de 1529 de setembronovembro de 2022.2024. Medical-deviceEstablishes riskthe classification,list notificationof equivalent foreign regulatory authorities and registration regimes, labelling, and instructionscriteria for use.optimized review under RDC 945/2024. AnvisaLegis. Accessed 19 July 2026.

7. AgenciaAgência Nacional de VigilanciaVigilância Sanitaria.Sanitária. ResolucaoResolução dade Diretoria Colegiada Anvisa- RDC No. 945,406, de 2922 de novembrojulho de 2024.2020. GuidelinesGood andpharmacovigilance procedurespractices for clinicalholders trialsof for laterhuman medicine registration.registrations. AnvisaLegis. Accessed 19 July 2026. Consolidated provisions checked 1 November 2026.

8. AgenciaAgência Nacional de VigilanciaVigilância Sanitaria.Sanitária. ResolucaoResolução - RDC No. 9,55, de 2016 de fevereirodezembro de 2015.2010. RegulationRegistration forof clinicalnew trialsbiological withproducts medicinesand inbiological Brazil.products. AnvisaLegis. Accessed 19 July 2026. Consolidated provisions checked 1 November 2026.

9. GaoAgência Y,Nacional XiongdeY,Vigilância GaoSanitária. X,Resolução et- al.RDC Retrieval-AugmentedNo. Generation751, de 15 de setembro de 2022. Medical-device risk classification, notification and registration regimes, labelling, and instructions for Largeuse. LanguageAnvisaLegis. Models:Accessed A19 Survey.July arXiv. 2023. doi:10.48550/arXiv.2312.10997.2026.

10. EsAgência S,Nacional Jamesde J,Vigilância Espinosa-AnkeSanitária. L,Resolução SchockaertdaS.Diretoria RAGAS:Colegiada Automated- EvaluationRDC No. 875, de 28 de maio de 2024. Complementary requirements for registration of Retrievalbiosimilars Augmentedthrough Generation.comparability. arXiv.AnvisaLegis. 2023.Accessed doi:10.48550/arXiv.2309.15217.23 October 2026.

11. ReimersAgência N,Nacional GurevychdeI.Vigilância Sentence-BERT:Sanitária. SentenceResolução Embeddings- usingRDC SiameseNo. BERT-Networks.948, In:de Proceedings12 de dezembro de 2024. General sanitary requirements for regularization of EMNLP-IJCNLP.human AssociationmedicinesforConsolidated ComputationaltextLinguistics;AnvisaLegis. 2019:3982-3992.Accessed doi:10.18653/v1/D19-1410.23 October 2026. Consolidated provisions checked 1 November 2026.

12. RobertsonEs S, ZaragozaJames H.J, TheEspinosa-Anke ProbabilisticL, RelevanceSchockaert Framework:S. BM25RAGAS: andAutomated Beyond.Evaluation FoundationsofandRetrieval TrendsAugmented inGeneration. InformationarXiv. Retrieval.2023. 2009;3(4):333-389. doi:10.1561/1500000019.doi:10.48550/arXiv.2309.15217.

13. BonifacioReimers L,N, JeronymoGurevych V,I. AbonizioSentence-BERT: HQ,Sentence CampiottiEmbeddings I,using FadaeeSiamese M,BERT-Networks. LotufoIn: R, Nogueira R. mMARCO: A Multilingual VersionProceedings of theEMNLP-IJCNLP. MSAssociation MARCOfor PassageComputational RankingLinguistics; Dataset.2019:3982-3992. arXiv. 2021. doi:10.48550/arXiv.2108.13897.doi:10.18653/v1/D19-1410.

14. LlamaIndex.Robertson LlamaIndex:S, aZaragoza dataH. frameworkThe forProbabilistic LLMRelevance applications.Framework: GitHubBM25 repository.and https://github.com/run-llama/llama_index.Beyond. AccessedFoundations 21and JulyTrends 2026.in Information Retrieval. 2009;3(4):333-389. doi:10.1561/1500000019.

15. GoogleBonifacio AIL, forJeronymo Developers.V, GeminiAbonizio 2.5HQ, Flash-LiteCampiotti modelI, documentation.Fadaee https://ai.google.dev/gemini-api/docs/models/gemini-2.5-flash-lite.M, AccessedLotufo 21R, JulyNogueira 2026.R. mMARCO: A Multilingual Version of the MS MARCO Passage Ranking Dataset. arXiv. 2021. doi:10.48550/arXiv.2108.13897.

16. ChangLlamaIndex. Y,LlamaIndex: WangaX,data WangframeworkJ,for etLLM al.applications. AGitHub Surveyrepositoryonhttps://github.com/run-llama/llama_index. EvaluationAccessed of21 LargeJuly Language Models. ACM Transactions on Intelligent Systems and Technology. 2024;15(3). doi:10.1145/3641289.2026.

17. ZhengGoogle L,AI ChiangforWL,Developers. ShengGemini Y,API et al. Judging LLM-as-a-Judge with MT-BenchmodelsandChatbotGenerateContentResponse Arena.reference. Advanceshttps://ai.google.dev/gemini-api/docs/models; inhttps://ai.google.dev/api/generate-content. NeuralAccessed Information24 ProcessingOctober Systems. 2023;36.2026.

18. Chang Y, Wang X, Wang J, et al. A Survey on Evaluation of Large Language Models. ACM Transactions on Intelligent Systems and Technology. 2024;15(3). doi:10.1145/3641289.

19. Zheng L, Chiang WL, Sheng Y, et al. Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena. Advances in Neural Information Processing Systems. 2023;36.

20. Sentence Transformers. all-MiniLM-L6-v2 model card. https://huggingface.co/sentence-transformers/all-MiniLM-L6-v2. Accessed 25 October 2026.

21. Sentence Transformers. Pretrained models and multilingual models. https://sbert.net/docs/sentence_transformer/pretrained_models.html. Accessed 25 October 2026.